

Elemental composition of lunar surface material (part 2)

A. A. SMALES, D. MAPPER, M. S. W. WEBB, R. K. WEBSTER, J. D. WILSON,
and J. S. HISLOP

Analytical Sciences Division, Atomic Energy Research Establishment, Harwell, England

(Received 9 February 1971; accepted in revised form 29 March 1971)

Abstract—Results are given for the elemental abundances in further samples of Apollo 11 materials and in three samples from the Apollo 12 mission. Comments are made on similarities and differences between Apollo 11 and Apollo 12 materials and between the latter and the meteorite Angra dos Reis.

INTRODUCTION

WE PRESENT HERE results so far obtained on samples 10057,52, 10059,34, 12022,49, 12065,51, and 12070,78, together with results for some further elements on 10060,26 and 10084,9,10. The methods used were activation analysis (AA); instrumental neutron activation analysis (INAA); radiochemical neutron activation analysis (RNAA); high-energy gamma activation analysis (GAA), and mass spectrometric isotope dilution analysis (MSID); supplemented by spark source mass spectrography (SSM) emission spectrography (ES), and X-ray fluorescence spectrometry (XRF). Their estimated accuracies are given in our previous paper (SMALES *et al.* 1970).

Results for minor and trace elements using AA and MSID are given in Table 1, those for major elements in Table 2, and some supplementary results by ES in Table 3. We have examined all our samples by SSM and have established upper limits for many more elements similar to those in Table 4 in our previous paper (SMALES *et al.* 1970).

DISCUSSION OF RESULTS

(1) The similarity of the two rocks 12022 and 12065, and the difference between them and the fines sample 12070 is readily apparent from Table 1. Apart from Sc, Cr, and Mn the fines sample is much higher in minor and trace element contents.

(2) The 12070 fines sample is significantly higher in K, Rb, Cs, Ba, REE, and Zr, but significantly lower in Ti and Zn, than the 10084 fines.

(3) Some of the values reported for the “volatile or depleted” elements of GANAPATHY *et al.* (1970) for types A and B rocks of Apollo 11 are given in Table 4 together with our results on the Apollo 12 rocks 12022 and 12065, listed by NASA (1970) as types B and AB, respectively. We have included As and Sb results by HASKIN *et al.* (1970) and a Ge result by WASSON and BAEDECKER (1970) on Apollo 11 rocks to compare with ours on Apollo 12, these elements being amongst the “normally depleted” elements of LARIMER (1967). It is clear that the pattern of behavior of all the elements listed is similar in the Apollo 11 and Apollo 12 rocks of the A and B types, though As seems significantly lower in the Apollo 12 rocks. It is perhaps worth recording here that our Ge, Ga, and In determinations on rock 12065 were by

Table 1. Minor and trace elements determined by activation analysis and/or mass spectrometric isotope dilution analysis.

Method*			Apollo 11		Apollo 12		
			10057	10059	12022	12065	12070
Parts per million by weight							
9	F	4	90	90	40	40	
11	Na	1	4040	3740	1680	1790	3690
19	K	5	2740	1600	510	520	2170
21	Sc	1	86	68	59	60	43
24	Cr	1	2250	2070	—	3400	2800
25	Mn	1	1810	1720	2410	2050	1750
27	Co	3	(~30 ²)		48.5	40	43
29	Cu	3				8	
30	Zn	3			2	2	11
37	Rb	3, 5	5.1, 6.3	3.2, 4.2	0.7, 0.8	1.0, 1.2	6.4, 6.6
38	Sr	4, 5	—, 190	—, 160	120, 170	110, 115	—, 190
47	Ag	3	0.006	0.009	0.011		0.026
55	Cs	3, 5	0.20, 0.21	0.12, —	0.04, 0.044	0.054, 0.06	0.29, 0.36
56	Ba	3, 5	260, 310	206, 210	54, 55	68, 67	390, 350
57	La	2	~28	~19	~5.5	~8	~38
58	Ce	5	(77 ²)	66	17	16	90
60	Nd	5	(63 ²)	51	15	14	57
62	Sm	1, 5	22, 21	15, 14	6, 5.4	5, 4.7	18, 16.7
63	Eu	1, 5	2.1, 1.9	2.1, 2.1	1.0, 1.3	0.9, 0.9	1.7, 1.7
65	Tb	2	~7	~4.5	~2.4	~2.3	~6
66	Dy	1	36	25	13	12	29
67	Ho	2	—	—	~3.1	~2.3	~6.7
70	Yb	1	19	13	4.9	4.7	13.6
71	Lu	2	~2.7	~1.9	~0.7	~0.6	~1.9
72	Hf	2	~17		~4	~4	~12

* Methods: (1) INAA $\pm$ 5%, (2) INAA $\pm$ 20%, (3) RNAA $\pm$ 5%, (4) GAA $\pm$ 10%, (5) MSID $\pm$ 5% where two methods are quoted results are given, respectively.
Additional information by method 3 on sample 12065: Ga 2.8, Ge 0.02, As 0.003, Pd <0.001, In 0.0024 Sb 0.009, Re <0.001, Au 0.0003 ppm. On sample 10084: Co 28.6, Rb 3.0, Cs 0.11, La 16, Zn 23. On Sample 10060: Co 27.4, Rb 4.2, La 21, Zn 27.

Table 2. Major elements in lunar rocks expressed as % oxide

Apollo 11							Apollo 12										
	XRF	10057	MSID	XRF	10059	MSID	10060	XRF	12022			GAA	XRF	12065			GAA
		INAA			INAA		XRF		INAA	MSID				INAA	MSID		
SiO ₂	41.4			42.4			41.8	42.3					47.1				
Al ₂ O ₃	8.1	7.6		12.6	12.5		11.5	8.2	8.5				9.8				
FeO	19.1			16.4			16.7	21.6			21.7		20.0				20.3
MgO	7.4			8.1			7.6	11.9					9.2				
CaO	10.4			11.5			11.2	9.0			9.2		10.1				10.9
Na ₂ O		0.54			0.51				0.23					0.24			
K ₂ O			0.33			0.19				0.061					0.063		
TiO ₂	10.9			8.3			8.7	4.9			4.8		3.1				3.4
MnO		0.23			0.22				0.31					0.26			
Cr ₂ O ₃		0.35			0.32									0.50			

Insufficient sample was available for XRF method on fines 12070. Results for Na, K, Mn, Cr, and trace elements on that sample are available in Table 1.

radio-chemical neutron activation on a single chip which was surface etched with acid after the irradiation, the etch liquor being discarded.

(4) In the fines, not only are the levels of alkali metals higher in 12070 than in 10084, but there has clearly been less differentiation of the heavier alkali metals in 12070, as can be seen from Table 5, where ratios are shown. Table 6 shows that the Apollo 12 rocks fit into the “low-Rb” classification of GAST *et al.* (1970) for Apollo

Table 3. Some supplementary results by emission spectrography*

			Apollo 11		Apollo 12		
			10057	10059	12022	12065	12070
Parts per million by weight							
28	Ni	< 20	300		50	20	200
23	V	50	50		120	150	110
29	Cu				5		
39	Y	250	190		60	45	130
40	Zr	770	630		200	180	600

* Estimated accuracy for ES. ± 20%.

Table 4. “Depleted” elements in Apollo 11 and Apollo 12 rocks.

	Apollo 11 Samples “Average” values for some type A and type B rocks*	Apollo 12 Samples Results for Rock 22 and/or Rock 65
Cu ppm	9.33	8.2
Zn ppm	7.41	2
Co ppm	20.9	48.5, 40
Pd ppb	3.40	< 1
As ppb	68†	3
Sb ppb	6.4‡	9
Ge ppm	≤0.08‡	0.02
Ga ppm	4.48	2.8
In ppb	2.9	2.4
Cs ppb	121	40–60
Rb ppm	3.58	0.8, 1.1

* From GANAPATHY *et al.* (1970).

† From HASKIN *et al.* (1970).

‡ From WASSON and BAEDECKER (1970).

Table 5. Alkali metal ratios in Apollo 11 and Apollo 12 fines.

	Apollo 11 Fines, 10084	Apollo 12 Fines, 12070
Na/K	2.67	1.70
Na/Rb	1067	568
Na/Cs	29,091	11,182
K/Rb	400	334
K/Cs	10,909	6,576
Rb/Cs	27.3	19.7

Table 6. Alkali metal contents of Apollo 11 and Apollo 12 rocks.

	Apollo 11 samples			Apollo 12 samples	
	10057, type A	10058*, type B	10060, type C	12022	12065
Na, ppm	4040	2900	3770	1680	1790
K	2740	877	1700	506	520
Rb	5.7	0.98	4.2	0.8	1.1
Cs	0.20	0.027	0.17	0.04	0.06

* GAST *et al.* (1970).

Table 7. Alkali metal ratios in Apollo 11 and Apollo 12 rocks.

	Apollo 11 samples			Apollo 12 samples	
	10057, type A	10058*, type B	10060, type C	12022	12065
Na/K	1.47	3.31	2.22	3.32	3.44
Na/Rb	709	2959	898	2100	1630
Na/Cs	20,200	107,400	22,180	42,000	29,830
K/Rb	481	895	405	633	473
K/Cs	13,700	32,480	10,000	12,650	8670
Rb/Cs	28.5	36.3	24.7	20.0	18.3

* Using results from GAST *et al.* (1970).

Table 8. Some REE ratios in Apollo 11 and Apollo 12 materials

	A	Apollo 11*			Apollo 12		
		B	C	D	12022	12065	12070
La/Sm	1.0	0.57	1.1	1.1	0.95	1.6	2.2
Sm/Eu	9.5	6.5	7.4	7.1	4.5	5.2	10.2
Lu/Sm	0.11	0.13	0.12	0.12	0.12	0.12	0.11
Yb/Lu	7.4	7.1	7.2	7.1	7.0	7.8	7.2

* From GAST *et al.* (1970), WAKITA *et al.* (1970).

Table 9. Elemental similarities between Angra dos Reis and Apollo 12 rocks.

	Angra dos Reis	12022	12065
Al%	5.7	4.4	5.2
Ti%	1.7	2.9	1.9
Sc ppm	57	59	60
Sr	133*	145	115
La	10	6	8
Ce	21†	17	16
Nd	20†	15	14
Sm	8	6	5
Eu	2.0	1.0	0.9
Tb	2	2	2
Dy	12	13	12
Ho	3.5	3	2
Yb	6	5	5
Lu	0.7	0.7	0.6

* TERA *et al.* (1970).

† SCHNETZLER and PHILPOTTS (1969).

11 rocks, generally of type B. However, the alkali metal ratios in Table 7 show that less differentiation of the heavier alkali metals has occurred in the Apollo 12 rocks, as compared with the Apollo 11 type B rocks. Indeed, in the Apollo 12 rocks, ratios between K, Rb, and Cs are nearer those for types A, C, and D of Apollo 11.

(5) As with Apollo 11 materials, there is a negative Eu anomaly in the Apollo 12 samples though not so pronounced in the rocks as in the fines. In Apollo 11 the REE levels were generally higher in type A than in type C, which in turn was higher than type B, with type C levels being similar to or higher than type D; we have only three samples of Apollo 12 material, but, as can be seen from Table 1, while the levels in 12070 (type D) are much higher than those for 12022 (type B), in line with

the Apollo 11 materials, 12070 is also much higher than 12065 (type AB). Table 8 lists some of the elemental abundance ratios used for example by GAST *et al.* (1970) and WAKITA *et al.* (1970) to discuss REE behavior in Apollo 11 materials, together with corresponding ratios from our Apollo 12 samples. The following points emerge:

(a) La/Sm: The value for 12070 (type D) is much higher than that for Apollo 11 type D; that for the 12022 (type B) rock is similar to those of Apollo 11 types A, C, D but much higher than that of Apollo 11 type B; while that for the 12065 (type AB) is higher than for any Apollo 11 type.

(b) Sm/Eu: The value for 12070 (type D) is much higher than for Apollo 11 types B, C, or D and even higher than for Apollo 11 type A, whereas the values for both Apollo 12 rocks are lower than any of the Apollo 11 types, even type B.

(c) Lu/Sm and Yb/Lu: The values in all three Apollo 12 samples are much the same as for all types of Apollo 11.

From these considerations it seems that there is some difference of behavior of the light REE in the Apollo 12 materials from that in Apollo 11.

(6) As was the case with Apollo 11 materials, we have compared the elemental abundances of the Apollo 12 samples with those available for the unique achondrite Angra dos Reis. It has not been our intention to suggest that Angra dos Reis might have derived from the moon, but rather that any similarity in patterns of elemental abundance might be used to assist in the thinking about the crystallization or melting processes involved in the lunar surface rock formation. Table 9 lists values from our previous paper (SMALES *et al.* 1970), for 14 elements in the meteorite and in the two Apollo 12 rocks available to us. The similarity is quite remarkable. However, the differences in levels of calcium and the alkali metals, noted in our previous paper, still persist for the Apollo 12 materials.

Acknowledgments—We thank M. W. Cooper, R. Hallett, T. C. Hughes, R. P. Kay, M. Perkins, A. G. Pratchett, C. A. J. McInnes, P. F. Ralph, and D. R. Williams for experimental assistance.

REFERENCES

- GANAPATHY R., KEAYS R. R., LAUL J. C., and ANDERS E. (1970) Trace elements in Apollo 11 lunar rocks: Implications for meteorite influx and origin of moon. *Proc. Apollo 11 Lunar Sci. Conf., Geochim. Cosmochim. Acta* Suppl. 1, Vol. 2, pp. 1117–1142. Pergamon.
- GAST P. W., HUBBARD N. J., and WIESMANN H. (1970). Chemical composition and petrogenesis of basalts from Tranquillity Base. *Proc. Apollo 11 Lunar Sci. Conf., Geochim. Cosmochim. Acta* Suppl. 1, Vol. 2, pp. 1143–1163. Pergamon.
- HASKIN L. A., ALLEN R. O., HELMKE P. A., PASTER T. P., ANDERSON M. R., KOROTEV R. L., and ZWEIFEL K. A. (1970) Rare earths and other trace elements in Apollo 11 lunar samples. *Proc. Apollo 11 Lunar Sci. Conf., Geochim. Cosmochim. Acta* Suppl. 1, Vol. 2, pp. 1213–1231. Pergamon.
- LARIMER J. W. (1967) Chemical fractionations in meteorites—I. Condensation of the elements. *Geochim. Cosmochim. Acta* **31**, 1215–1238.
- NASA (1970) Apollo 12 preliminary science report. NASA SP-235. Houston.
- SCHNETZLER C. C., and PHILPOTTS J. A. (1969) Genesis of the calcium-rich achondrites in light of rare earth and barium concentrations. In *Meteorite Research* (editor P. M. Millman), pp. 206–216. Reidel.
- SMALES A. A., MAPPER D., WEBB M. S. W., WEBSTER R. K., and WILSON J. D. (1970) Elemental composition of lunar surface material. *Science* **167**, 509–512.
- TERA F., EUGSTER O., BURNETT D. S., and WASSERBURG G. J. (1970) Comparative study of Li, Na, K, Rb, Cs, Ca, Sr, and Ba abundances in achondrites and in Apollo 11 lunar samples.

- Proc. Apollo 11 Lunar Sci. Conf., Geochim. Cosmochim. Acta* Suppl. 1, Vol. 2, pp. 1637–1657. Pergamon.
- WAKITA H., SCHMITT R. A., and REY P. (1970) Elemental abundances of major, minor, and trace elements in Apollo 11 lunar rocks, soil, and core samples. *Proc. Apollo 11 Lunar Sci. Conf., Geochim. Cosmochim. Acta* Suppl. 1, Vol. 2, pp. 1685–1717. Pergamon.
- WASSON J. T., and BAEDECKER P. A. (1970) Ga, Ge, In, Ir, and Au in lunar, terrestrial, and meteoritic basalts. *Proc. Apollo 11 Lunar Sci. Conf., Geochim. Cosmochim. Acta* Suppl. 1, Vol. 2, pp. 1741–1750. Pergamon.